SCC Utility CLI User Guide

This guide provides an in-depth overview of the **scc_utility** command-line interface (CLI), which automates common Security Cloud Control (SCC) tasks such as deploying and upgrading Firepower Threat Defense (FTD) devices, and monitoring upgrade transactions.

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Overview

The **scc_utility** CLI simplifies and centralizes operations against the Security Cloud Control (SCC) APIs:

- **Deploy** pending configuration changes to one or more FTD devices
- **List** available and compatible firmware versions
- **Upgrade** FTDs individually or in bulk
- **Monitor** upgrade transactions in real time

All commands share a common entry point (`app.py`), invoking subcommands for each operation.

Prerequisites

- **Windows OS** (Windows 10 or later)
- Access to the Netteam-Globalconnect GitHub organization (invite required)
- Valid SCC **API token** with permissions to query inventory and perform deployments/upgrades

Installation

1. **Download** the latest MSI installer from the SCC Utility GitHub Releases page (requires a GitHub account and membership invitation): https://github.com/Netteam-Globalconnect/scc_utility/releases/latest
2. **Run** the downloaded `.msi` file and follow the on-screen prompts to install the application.
3. After installation, the `scc_utility` CLI tool will be available in your Start menu or via the command prompt.

Authentication and Configuration

MSP Management

For MSP (Managed Service Provider) users, the CLI offers dedicated tenant management workflows:

1. **Tenant Selection**: On startup, if your API token belongs to an MSP account, you'll be prompted to select one of your managed tenants from the MSP portal.

2. **API-Only User Provisioning**: The CLI ensures an API-only user exists in the selected tenant and, if not, provisions one automatically.
3. **Tenant-Scoped Token Generation**: A fresh, tenant-scoped API token is generated for that API-only user and used for all subsequent operations.
4. **Seamless Context**: After selection, all commands (`deploy`, `upgrade`, etc.) run in the context of the chosen managed tenant.

If no managed tenants are found, the CLI falls back to using your original API token and operates at the root level.

Before executing any commands, you must supply your SCC API credentials. You can provide them via:

- **Command-line options** (`--region`, `--api-token`)
- **Interactive prompt** (you'll be prompted if neither option is set)

Credentials are managed by the `SccCredentialsService`, which caches or prompts to ensure secure handling.

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Global Options

All subcommands inherit these global options:

Option	Description	Required	Example
`--region`	SCC region identifier	No*	`eu`
`--api-token`	SCC API access token (string)	No*	`ABCDEF12345`

*If omitted, you will be prompted at runtime or credentials loaded from cache.

Command Reference

Below is a detailed breakdown of each available subcommand.

`deploy`

Syntax:

```
scc_utility deploy [OPTIONS]
```

Description:
Deploys any pending configuration changes to a single FTD device.

Options:

Option	Description	Required
`--ftd-uid`	UID of the target FTD	No
`--ftd-name`	Name filter for selecting the FTD (prefix match)	No

Behavior:
1. If `--ftd-name` is provided and matches exactly one device, its UID is used.
2. If multiple devices match or neither `--ftd-uid` nor `--ftd-name` given, an interactive selector appears.
3. Deploys with default notes: `"Deployment via CLI"`.
4. Prints a success message on completion.

Exit Code:
- `0` on success
- Non-zero on failure

`deploy-bulk`

Syntax:

```
scc_utility deploy-bulk [OPTIONS]
```

Description:
Deploys pending changes to multiple FTD devices in bulk.

Options:
- *(None)* – all selection is interactive

Behavior:
1. Prompts to multi-select up to all online FTDs.
2. Deploys each device with a 1-second delay between calls.
3. Uses notes: `"Bulk deploy from CLI"`.

`list-versions`

Syntax:

`scc_utility list-versions [OPTIONS]`

Description:
Lists all firmware versions compatible with a given FTD.

Options:

Option	Description	Required
`--ftd-uid`	UID of the FTD	No
`--ftd-name`	Name of the FTD (exact match)	No

Behavior:
1. Resolves FTD via name or UID, or interactive selector if unspecified.
2. Fetches compatible versions via the `DeviceUpgradeApiService`.
3. Displays a table of versions, marking suggested versions with `*`.

`upgrade`

Syntax:

`scc_utility upgrade [OPTIONS]`

Description:
Upgrades the firmware on a single FTD to a specified or selected version.

Options:

Option	Description	Required
`--ftd-uid`	UID of the target FTD	No
`--ftd-name`	Name filter for selecting the FTD	No
`--upgrade-package-uid`	UID of the upgrade package (firmware image)	No

Behavior:
1. Resolves target FTD via name or UID; interactive if ambiguous.
2. If no upgrade package provided, prompts to select from compatible versions (suggested marked `*`).
3. Optionally deploys pending changes first (confirmation prompt).

4. Prompts for final confirmation before starting upgrade.
5. Executes the upgrade and prints status.

Interactive Prompts:
- Select firmware version
- Confirm pre-upgrade deploy
- Confirm upgrade

`upgrade-bulk`

Syntax:

```
scc_utility upgrade-bulk [OPTIONS]
```

Description:
Upgrades multiple FTD devices in bulk.

Options:

Option	Description	Required
`--upgrade-package-uid`	UID of the firmware package to apply	No

Behavior:
1. If no package UID provided, interactive version selection.
2. Prompts to multi-select up to 10 online FTDs.
3. Upgrades all selected devices in parallel.

`monitor-upgrade`

Syntax:

```
scc_utility monitor-upgrade --transaction-id <ID>
```

Description:
Continuously polls and displays the status of a bulk upgrade transaction.

Options:

Option	Description	Required
`--transaction-id`	ID of the transaction to monitor	Yes

Behavior:
1. Polls every 10 seconds by default.
2. Renders a live-updating table of device states until completion.

Interactive Mode

Whenever a command cannot resolve a device or firmware version from provided flags, you will enter an **interactive** mode powered by questionary and rich. Use arrow keys and spacebar to navigate and select items.

Examples

Deploy to a specific device by UID:

```
scc_utility deploy --ftd-uid a1b2c3d4 --region us-east-1 --api-token $SCC_TOKEN
```

Bulk upgrade with interactive version selection:

```
scc_utility upgrade-bulk --region eu-west-1
```

Monitor an upgrade transaction:

```
scc_utility monitor-upgrade --transaction-id tx-1234567890
```

Troubleshooting

- **Authentication errors**: Ensure your `--api-token` is valid and not expired. Re-run the command without `--api-token` to be prompted for fresh credentials.

- **Region not supported**: Verify the region code is one of the supported regions in `scc_utility/utils/region_mapping.py`.

- **No devices found**: Check that your SCC tenant has devices in an **ONLINE** state.

- **Bulk operations hanging**: Use `monitor-upgrade` to inspect progress or failures.

Best Practices

- Always run the `deploy` command before upgrading to ensure configuration consistency.

- Review compatible versions with `list-versions` to confirm upgrade paths.

- Use bulk commands for routine maintenance windows to minimize manual effort.

- Automate monitoring by piping `monitor-upgrade` output into logs or alerting systems.

For further assistance, consult the project repository or contact the SCC support team.